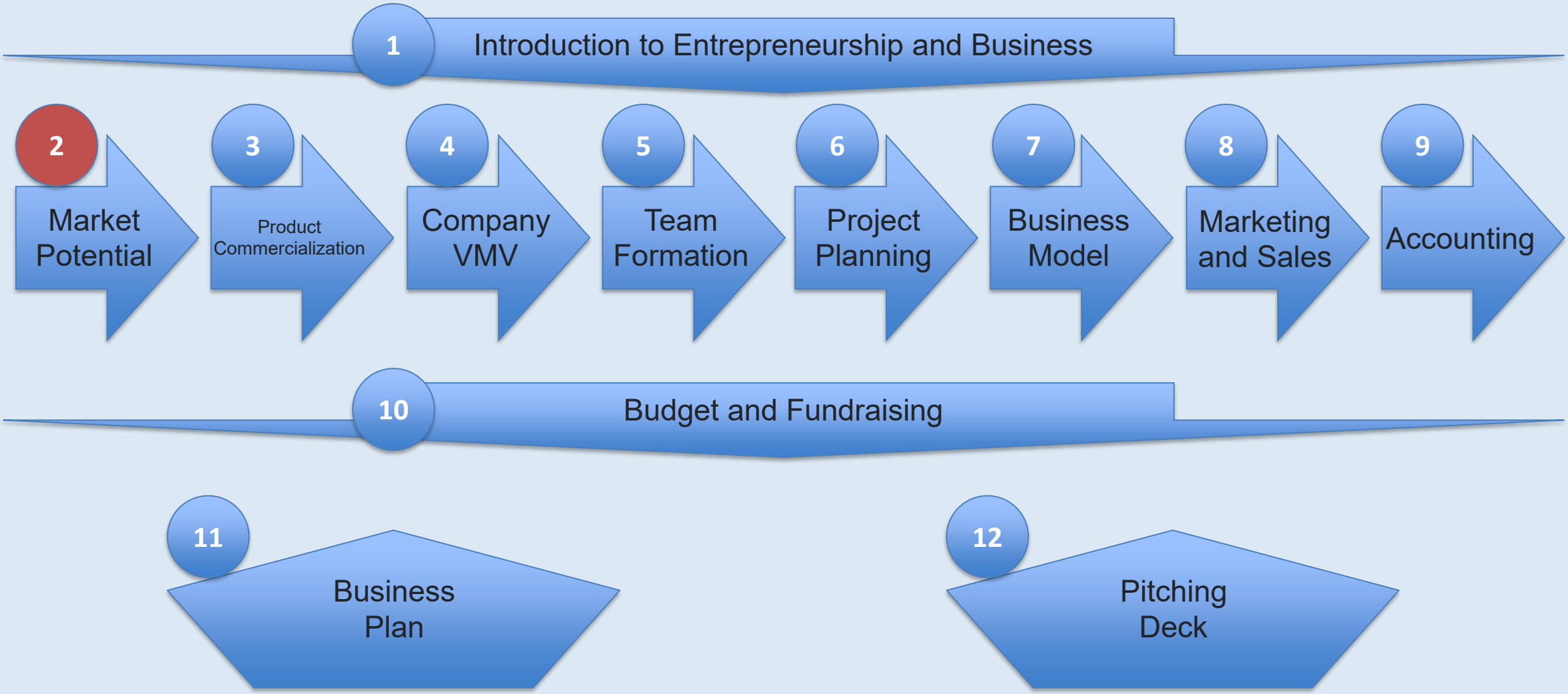
Market Potential Identification Methodologies

Ir Prof Alan Lam, JP



Powered by **Poffices.AI**

Outline of the Course



Chapter 1: Market Analysis and Competitive Analysis

Ir Prof Alan Lam, JP

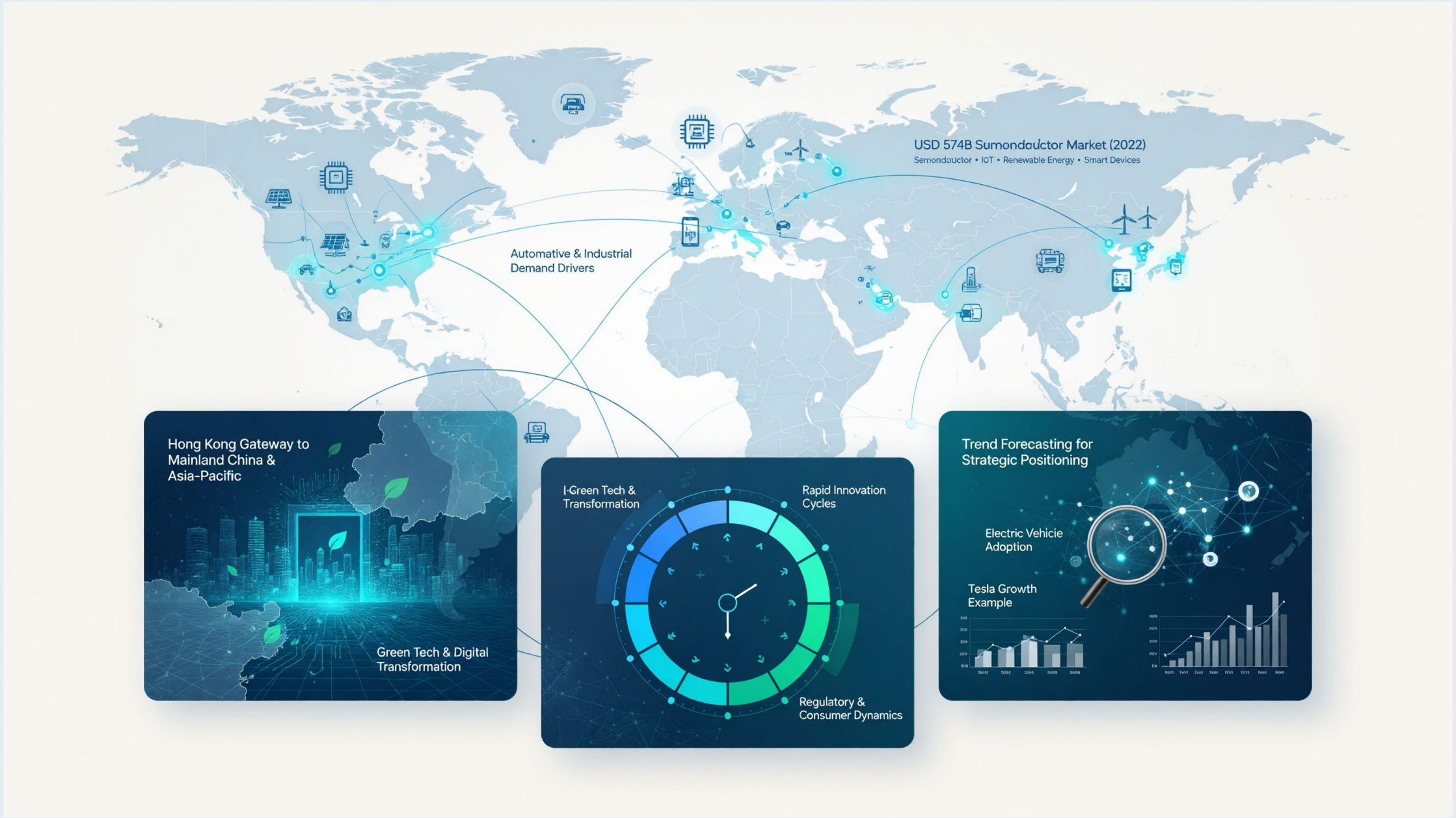


Powered by Poffices.AI

Market Overview and Dynamics



Industry Trends



Identification of Market Segments



Analyzing Market Share



Example of Market Segments

Assume the global TAM for smart sensor modules is HKD 30 billion. If the entrepreneur targets industrial automation applications representing 40% of TAM, then:

$$SAM = TAM \times 0.4 = 30 \text{ billion} \times 0.4$$

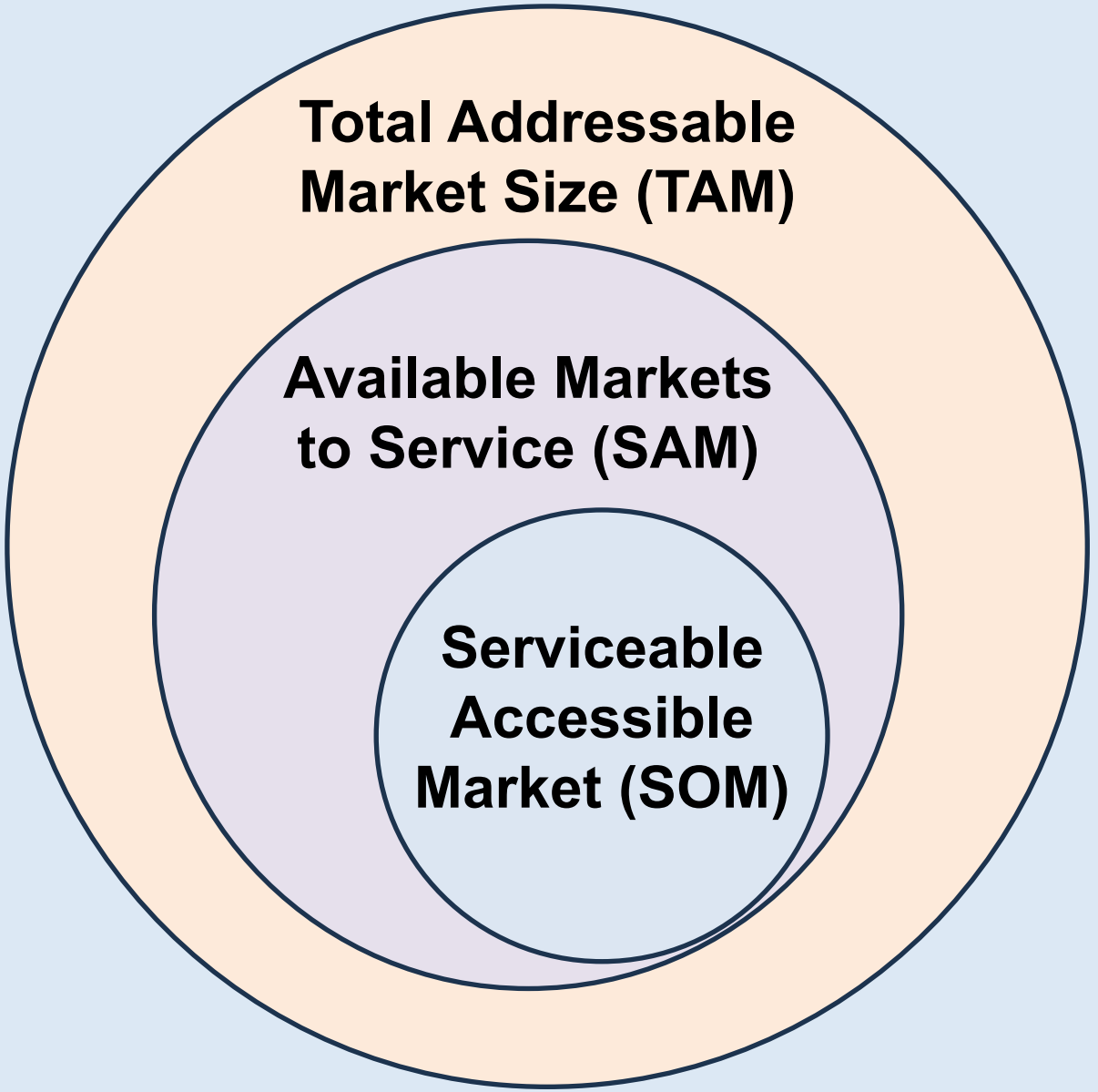
$$= 12 \text{ billion HKD}$$

If market entry barriers and current capabilities limit the reachable share to 10% of SAM, then:

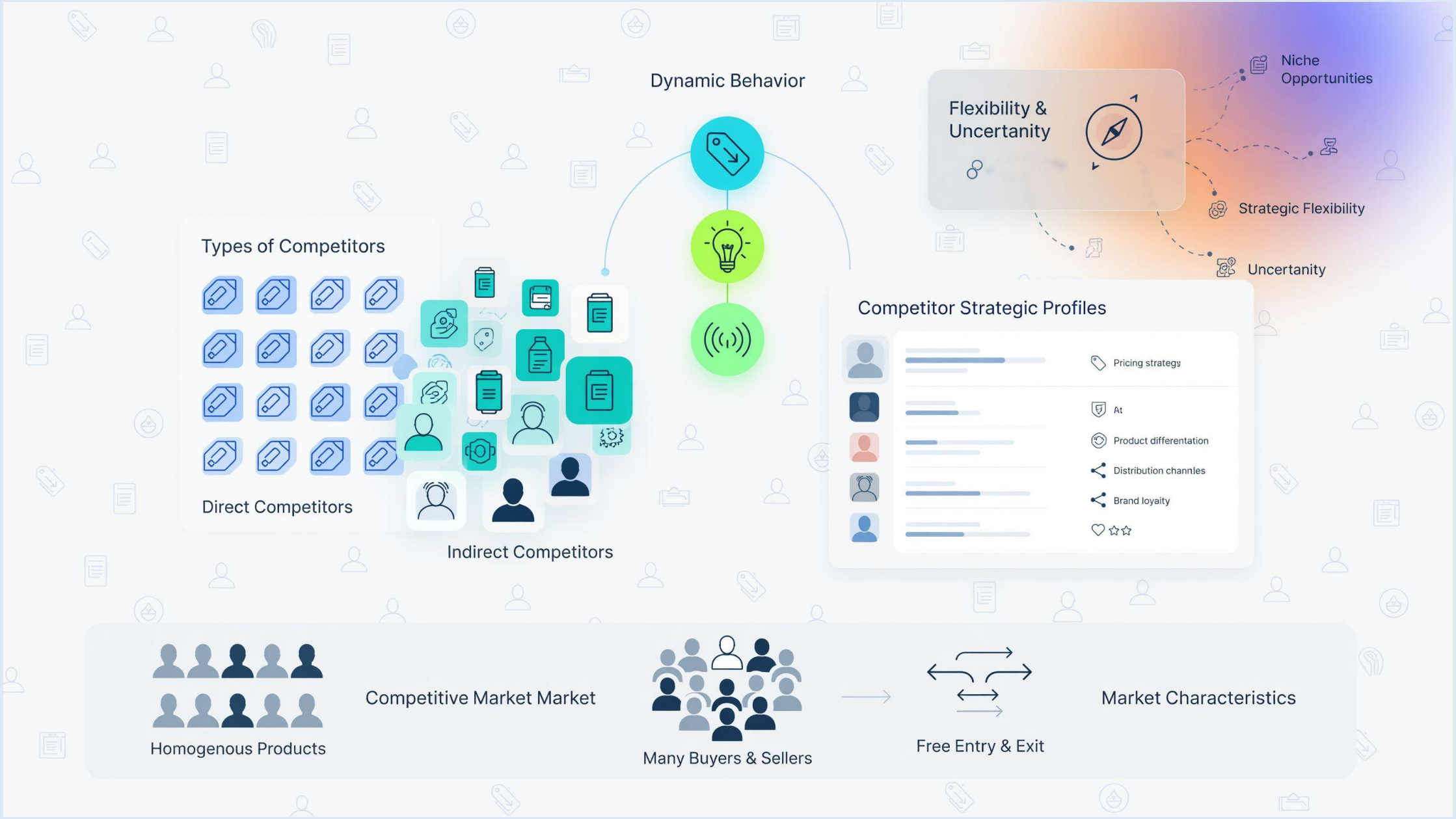
$$SOM = SAM \times 0.1 = 12 \text{ billion} \times 0.1$$

$$= 1.2 \text{ billion HKD}$$

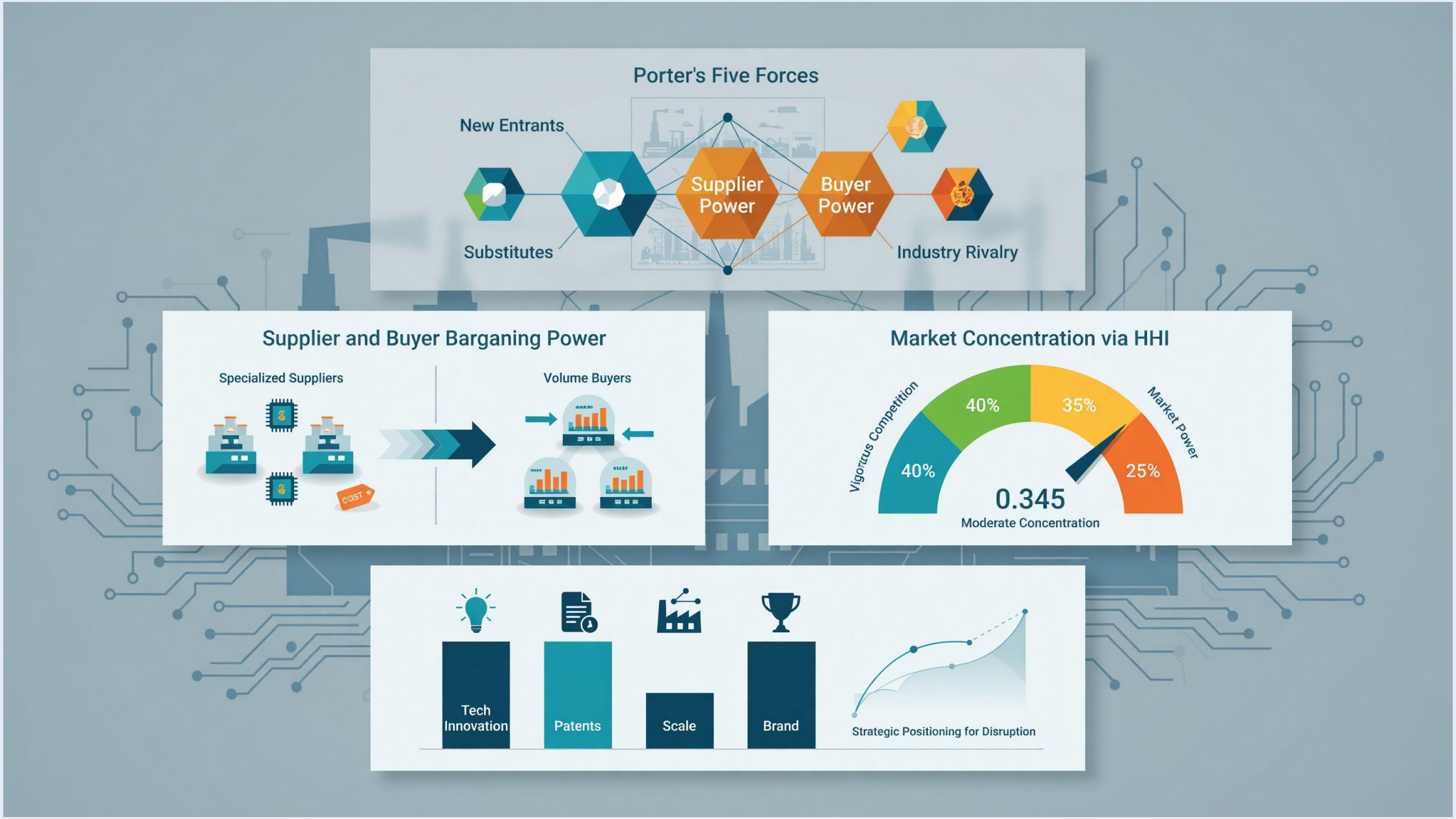
This step-by-step calculation defines realistic revenue expectations and guides resource allocation for market penetration efforts.



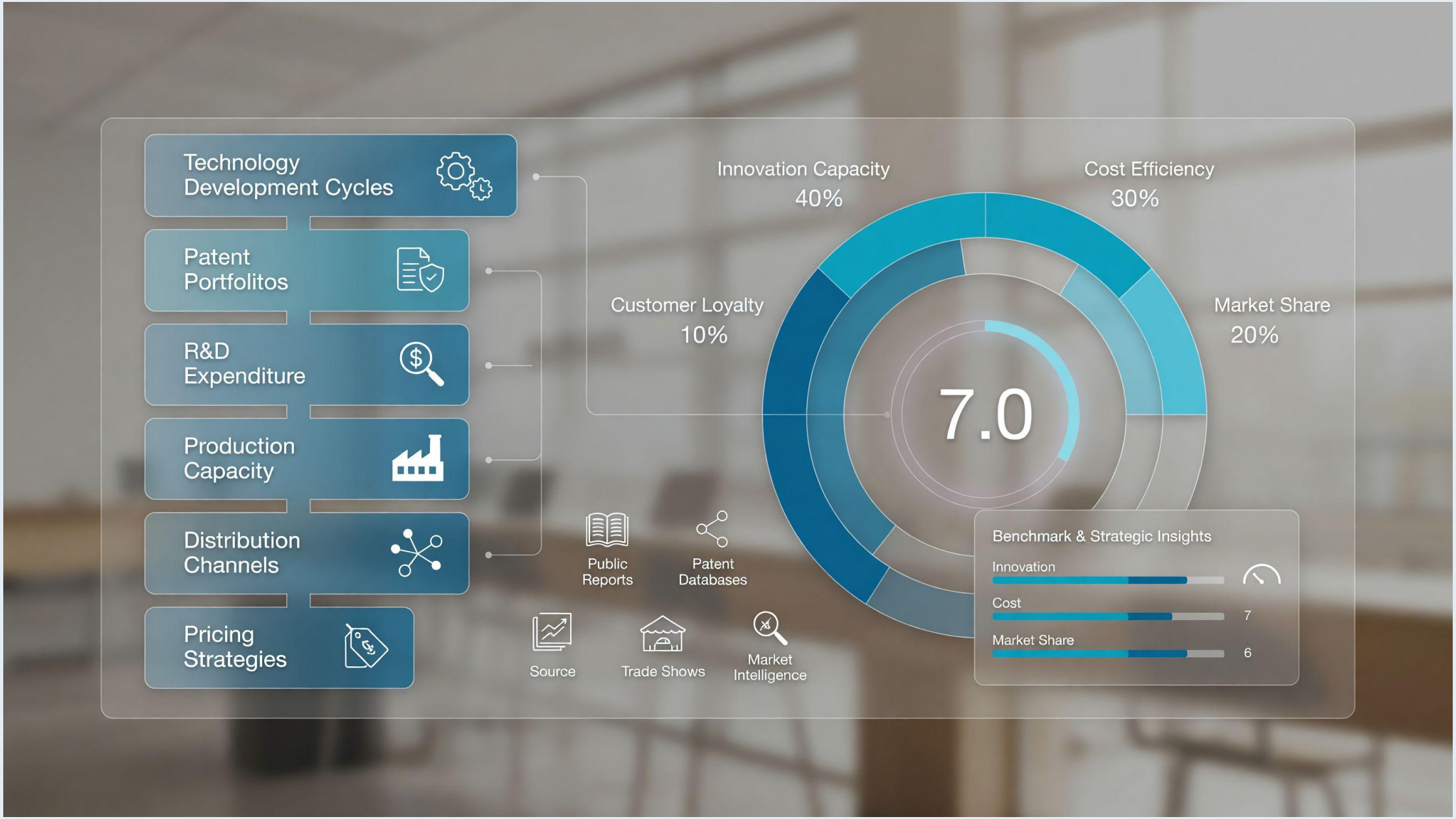
Understanding Competitors



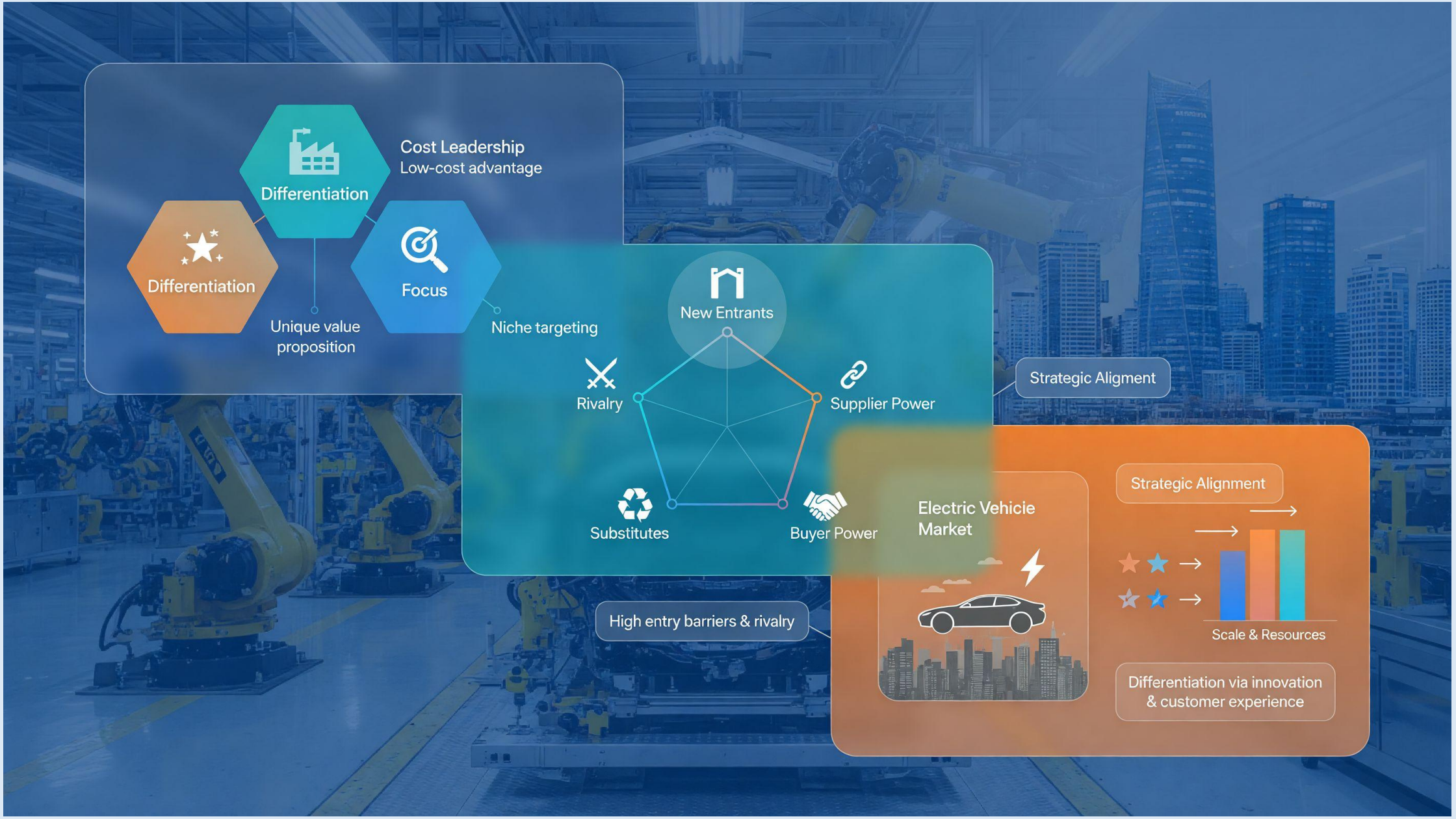
Framework for Competitive Analysis



Competitor Profiling Techniques

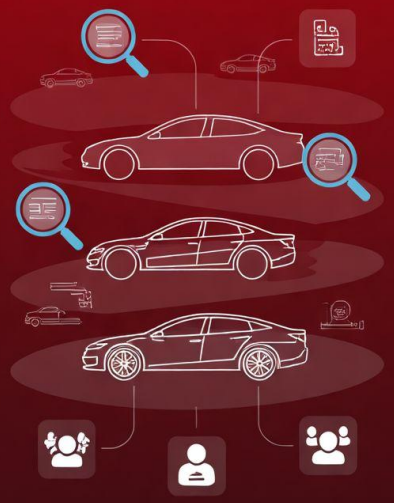


Competitive Strategy Analysis



Case Study: Competitive Analysis in Technology Market

Strategic Competitive Analysis



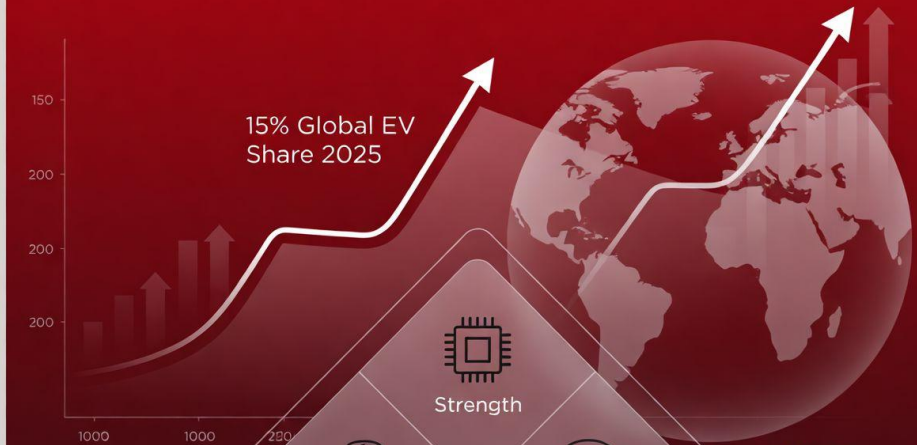
Competitive Gaps

Integrated Innovation and Market Positioning



Software & Battery Integration
Rapid Innovation Cycles
Vertical Supply Chain
Premium Branding

Market Success Evidenced by Growth



SWOT Insights

Strength
Weakness
Opportunity
Threat



Entrepreneur Growth Lessons

Practical Examples of Market and Competitive Analysis



Consider a startup developing IoT sensor modules targeting industrial automation in Hong Kong. The first step is market sizing using TAM/SAM/SOM calculations. Assume total IoT device market TAM is approximately HKD 50 billion regionally.

Targeting industrial automation yielding 30% of TAM results in SAM of:

$$SAM = 50\,\text{billion} \times 0.3 = 15\,\text{billion HKD}$$

Estimating realistic market share at 5% due to competitive intensity gives SOM:

$$SOM = 15\,\text{billion} \times 0.05 = 0.75\,\text{billion HKD}$$

Next, analyzing competitors reveals dominant firms with 40% market share controlling supply chain access, indicating the startup's need to innovate beyond price competition.

Applying Porter's Five Forces quantitatively:

- Rivalry intensity: High, with over 10 competitors.
- Threat of new entrants: Moderate, due to high capital requirements.
- Supplier power: High for unique components.
- Buyer power: Moderate; industrial customers demand customization.
- Substitutes: Emerging wireless sensor solutions.

Employing these analyses guides strategic decisions such as product differentiation and alliances with suppliers to mitigate risks and capitalize on market opportunities.

Summary of Key Analytical Tools



Chapter 2: Market Research Techniques

Ir Prof Alan Lam, JP

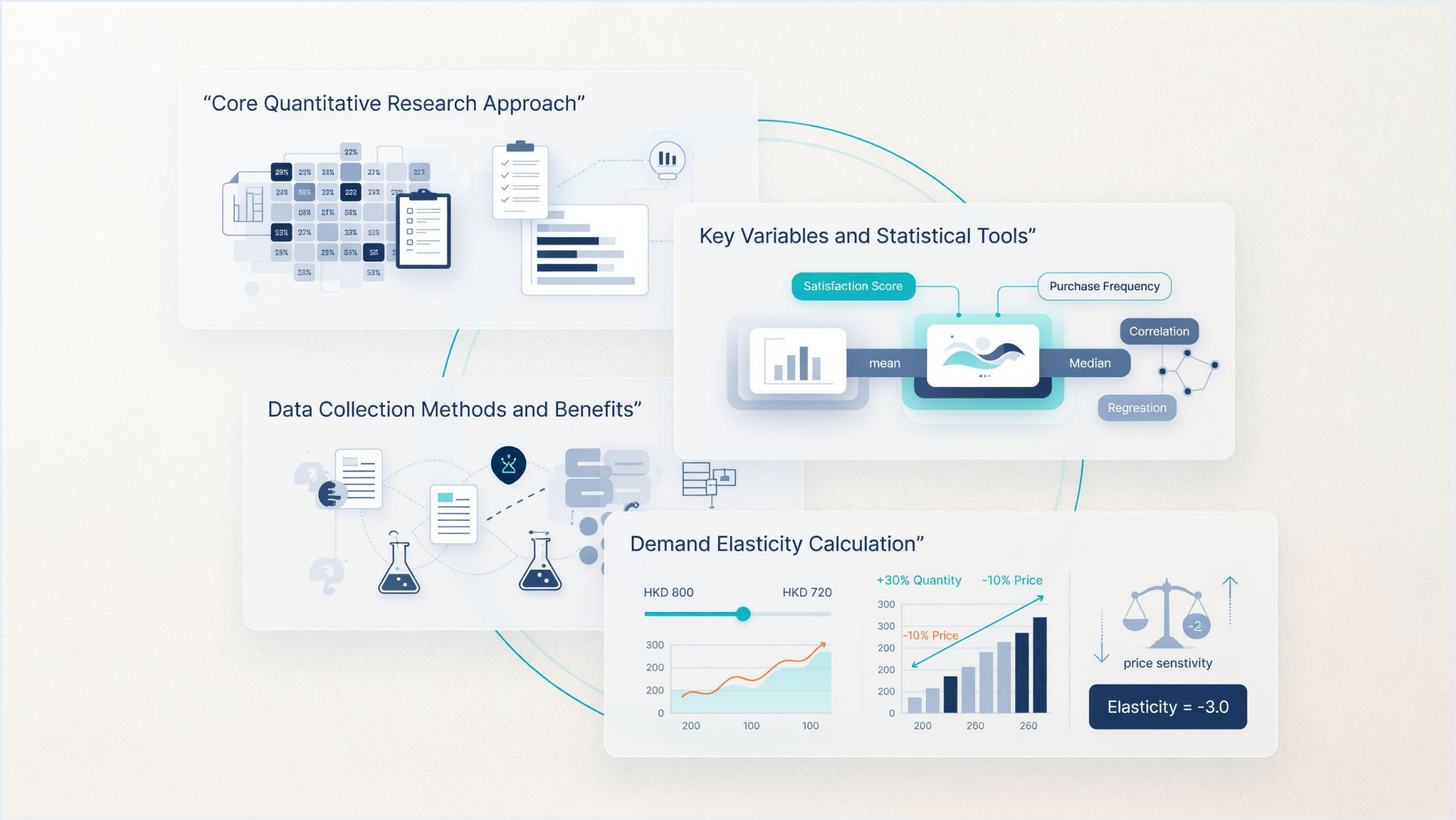


Powered by Poffices.AI

Qualitative Research Methods



Quantitative Research Methods



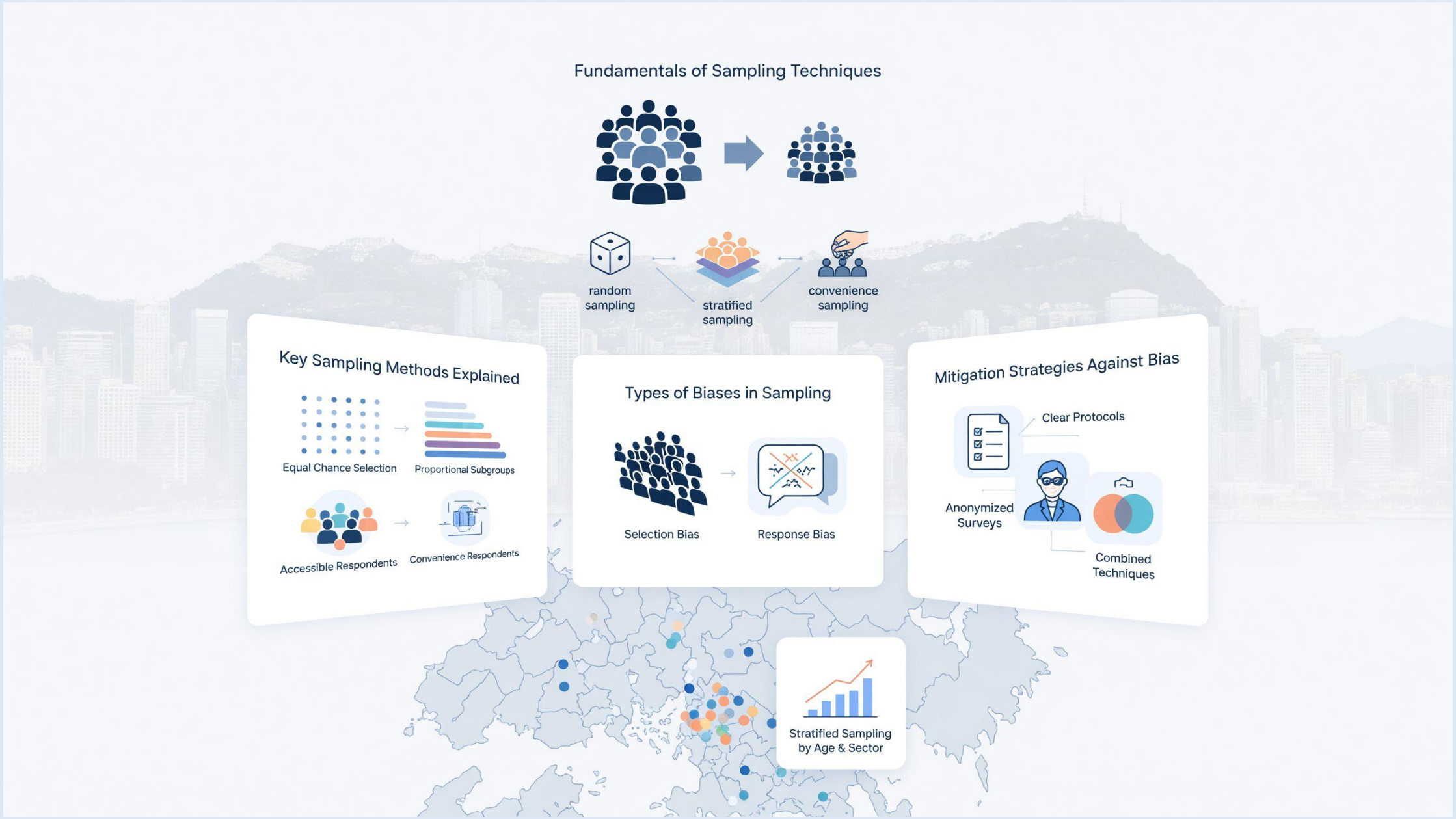
Quantitative Research Methods



Qualitative Research Methods



Sampling Techniques and Biases



Data Collection Techniques



Survey Design and Distribution

1



Interviewing Techniques

2





Observation with data Sampling and Bias Considerations

3

Role of Sampling in Research

Representative samples → generalizable conclusions
Biased samples → distorted results

Common Sampling Methods

- Random Sampling
- Stratified Sampling
- Convenience Sampling
- Snowball Sampling

Sources and Impact of Bias



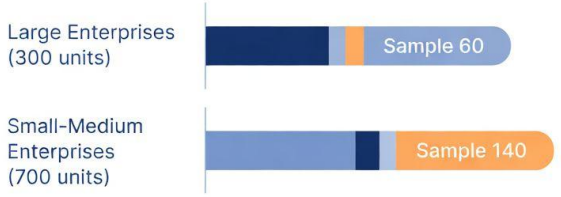
Case Study: IoT Manufacturer Survey



Bias Mitigation Techniques

- Pilot Testing
- Follow-up
- Weighting Adjustments

Stratified Sampling Allocation Explained

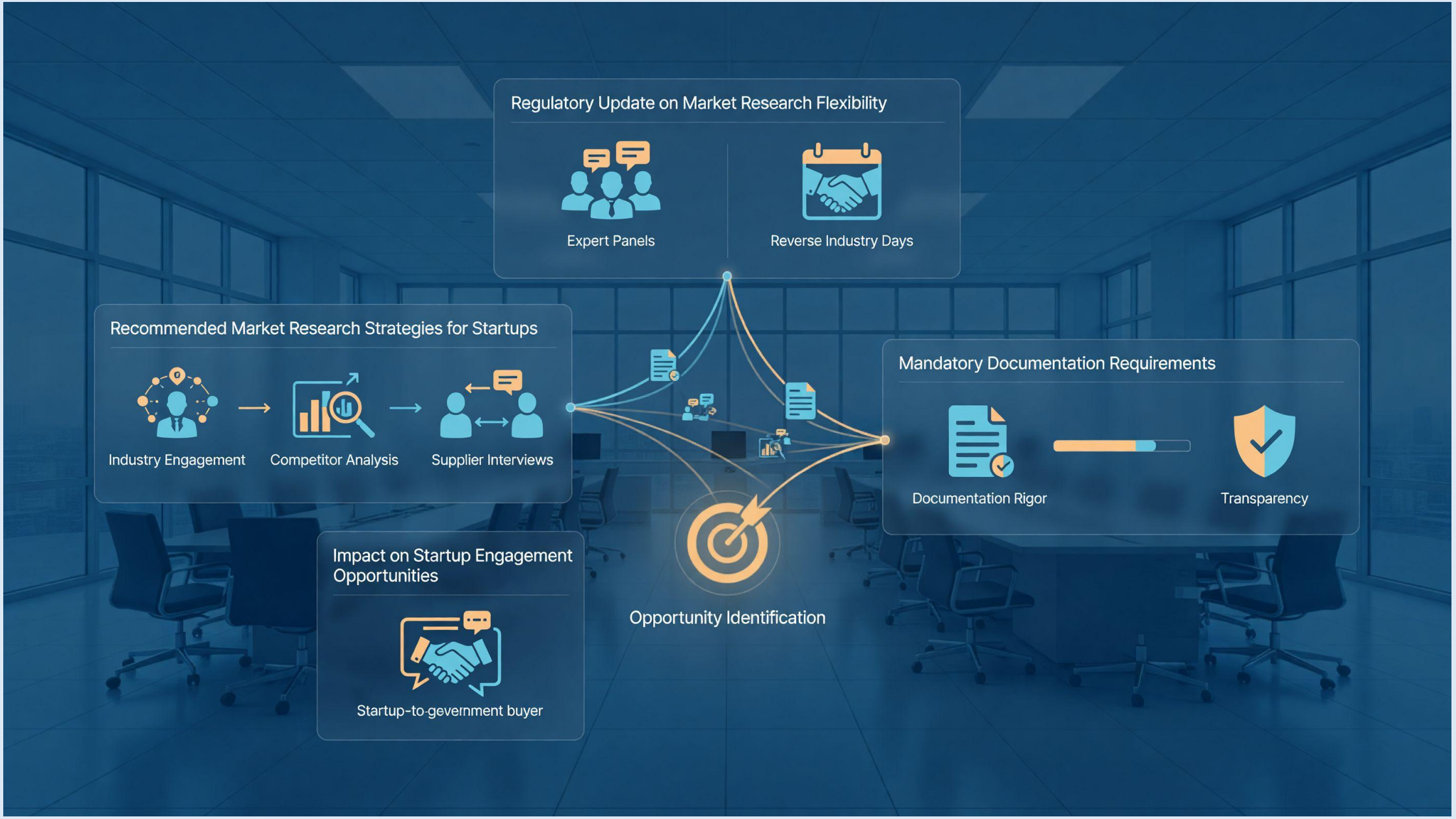


Data Analysis and Interpretation

4



Effective Research Implementation



Examples of Market Research in Technology Sectors

Semiconductor Startup



Power-efficient chip demand

HKD 2 billion TAM.

Industrial IoT Platform



Stratified sampling & regression

Adoption predictions by firm size & maturity.

Smart Home Device Company



Privacy concerns limit adoption

Security feature improvements.

Chapter 3: Market Size Calculation Methods (TAM/SAM/SOM)

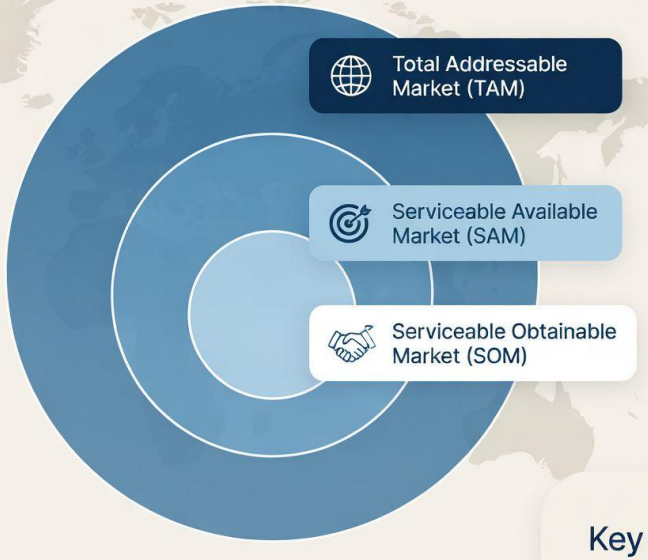
Ir Prof Alan Lam, JP



Powered by Poffices.AI

Introduction to Market Size Concepts





Key Market Size Metrics: TAM, SAM, SOM



Strategic Impact and Risks



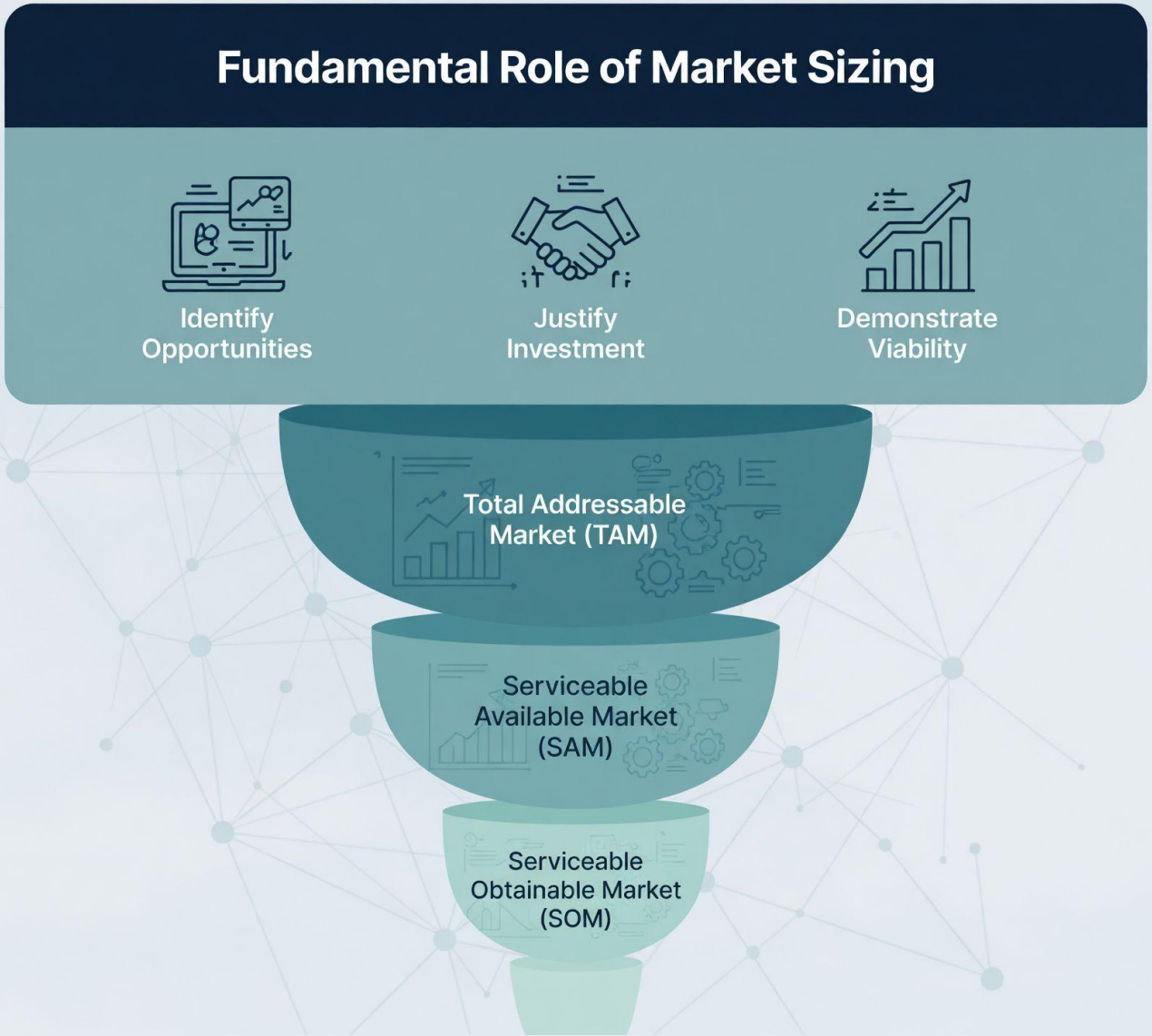
Credibility risk



Preparedness risk

Accurate Market Sizing Enables Strategy & Investment Confidence

Importance of Market Sizing Understanding

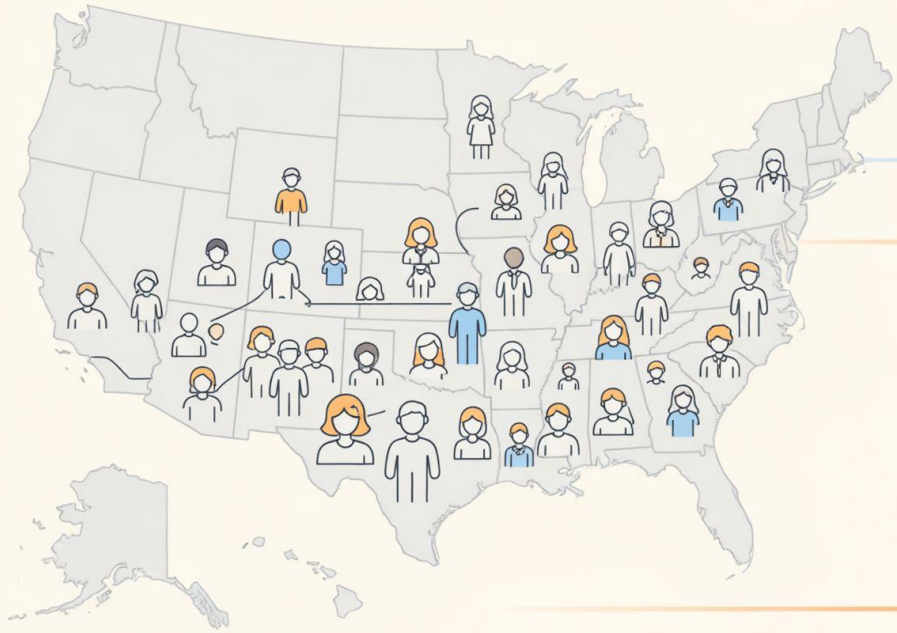


Total Addressable Market (TAM) Introduction



Calculation of Total Addressable Market (TAM)

Total Addressable Market (TAM): Maximum revenue potential at 100% market penetration





Industry Data → Product Market Segmentation



140M

Potential Customers

×



HKD 4,000

Annual Revenue per Customer



Company Data → Revenue Forecast

TAM = HKD 560 Billion

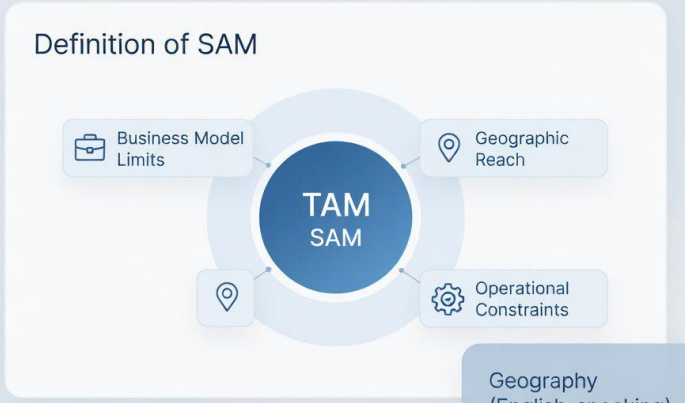
Serviceable Available Market (SAM) Introduction



Serviceable Obtainable Market (SOM) Estimation



Serviceable Available Market (SAM) Introduction



Geography (English-speaking)

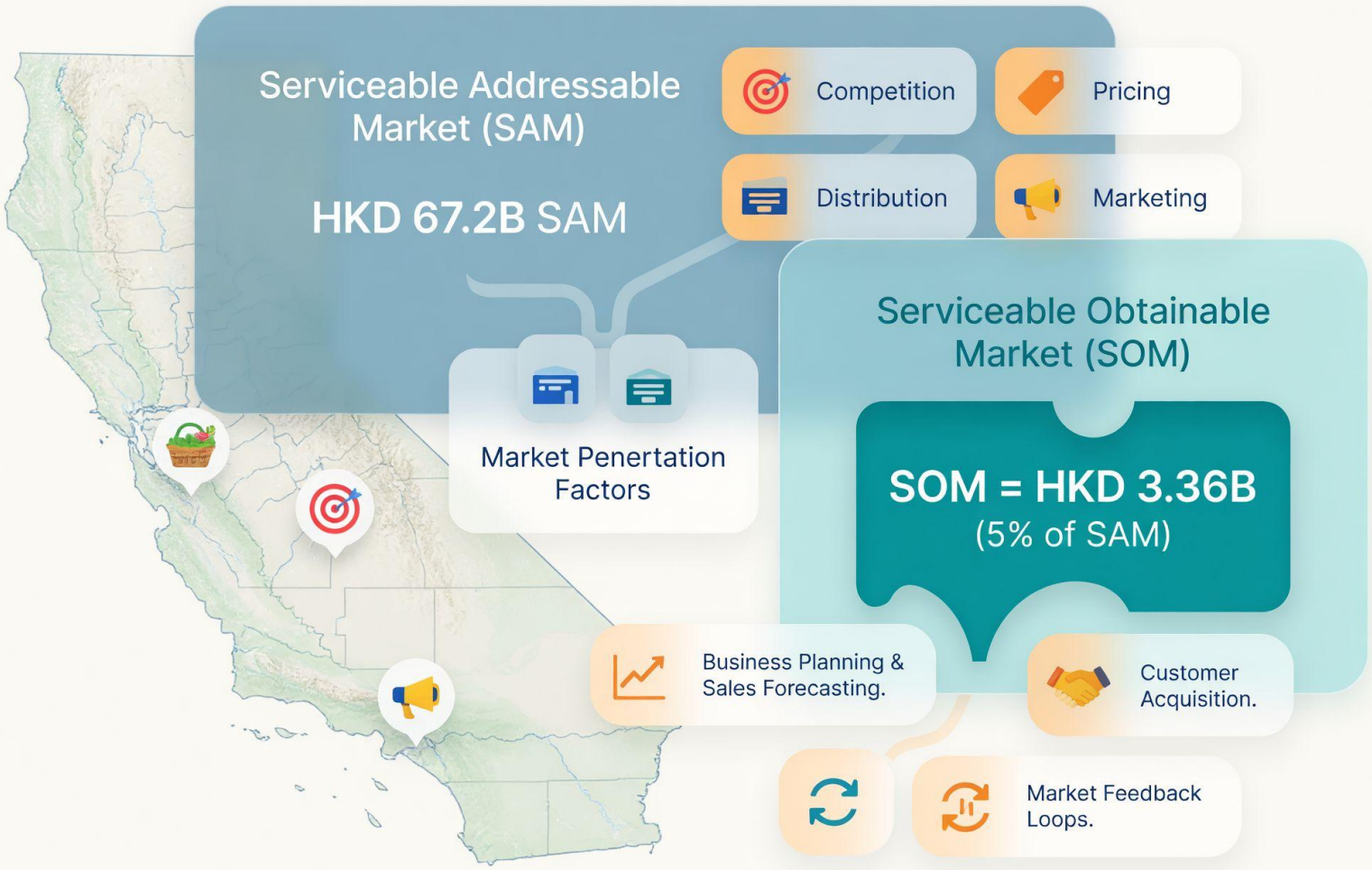
Customer Demographics

Product Applicability





Share of Market (SOM) Analysis



5%
Market Share

4-5
Year Horizon

Mathematical Formulas for TAM/SAM/SOM



The market size metrics TAM, SAM, and SOM can be expressed mathematically. Understanding these formulas supports accurate and transparent calculations.

Let:

- N_{total} = Total possible customers in the relevant market
- P = Price per unit or average revenue per customer annually
- F_{TAM} = Fraction of total customers in the overall market (usually 1 for TAM)
- F_{SAM} = Fraction of TAM served by company's product/service capabilities and reach
- F_{SOM} = Fraction of SAM realistically obtainable

Then,

$$\text{TAM} = N_{total} \times P \times F_{TAM}$$
$$\text{SAM} = \text{TAM} \times F_{SAM}$$
$$\text{SOM} = \text{SAM} \times F_{SOM}$$

Alternatively, if starting from unit volume:

$$\text{Market Size} = \text{Number of Units Sold} \times \text{Price per Unit}$$

In the bottom-up model, the number of units can be estimated by:

$$\text{Number of Units Sold} = \text{Number of Customers} \times \text{Average Units Bought per Customer}$$

Each fraction F_{SAM} and F_{SOM} must be supported by market research data or validated assumptions about reach and competitive share.

Step-by-step Numerical Examples of Market Size Calculations

To clarify these concepts, consider a startup developing a wearable heart rate monitor targeting the Hong Kong market, an example aligned with technology applications in Electrical and Electronic Engineering.

Step 1: Calculate TAM

- Total population of Hong Kong: approximately 7.5 million.
- Target customers: Adults aged 18-65 = approx. 5 million.
- Estimated percentage interested in health monitoring devices = 60%.
- Potential customers = 5,000,000 x 0.60 = 3,000,000.
- Average price per device = HKD 2,000.

Therefore,

$$\text{TAM} = 3{,}000{,}000 \times 2{,}000 = HKD6{,}000{,}000{,}000$$

Step 2: Estimate SAM

- The startup plans to initially serve urban areas with tech-savvy population which accounts for 50% of the TAM segment.
- Hence,
  $$\text{SAM} = \text{TAM} \times 0.50 = HKD6{,}000{,}000{,}000 \times 0.50 = HKD3{,}000{,}000{,}000$$

Step 3: Estimate SOM

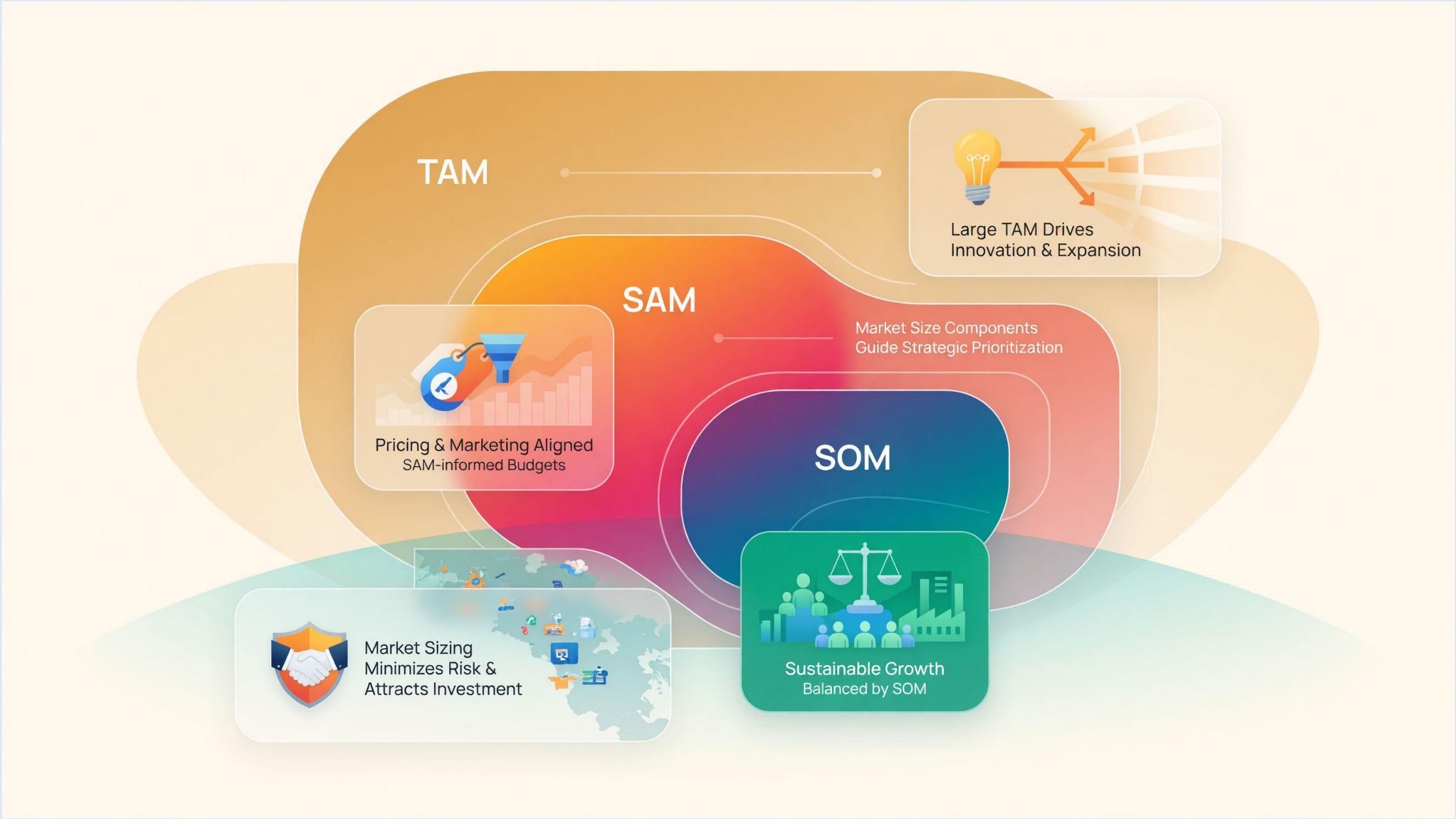
- Realistically expecting to capture 2% of SAM in the next 3 years based on marketing and sales capabilities.
- Thus,
  $$\text{SOM} = \text{SAM} \times 0.02 = HKD3{,}000{,}000{,}000 \times 0.02 = HKD60{,}000{,}000$$



Interpretation of Market Sizing Results for Strategic Decisions



Market Size Impact on Startup Strategy



Chapter 4: Market Analysis Methods (SWOT, PEST, Porter's 5 Forces)

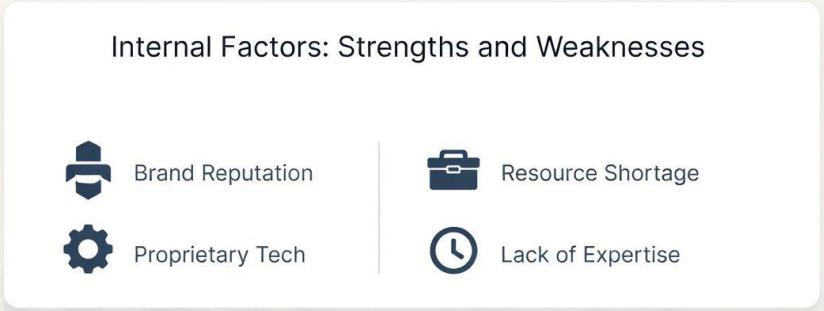
Ir Prof Alan Lam, JP

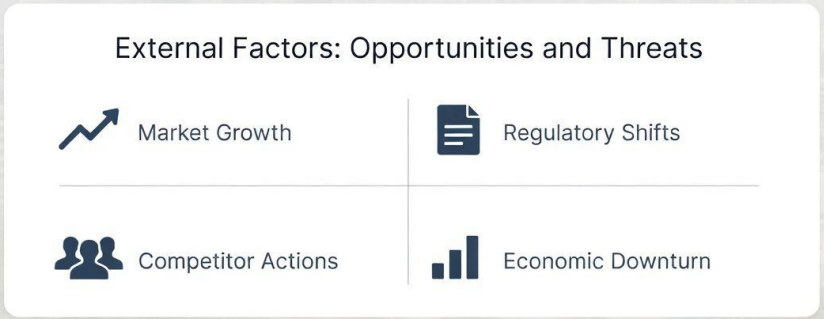


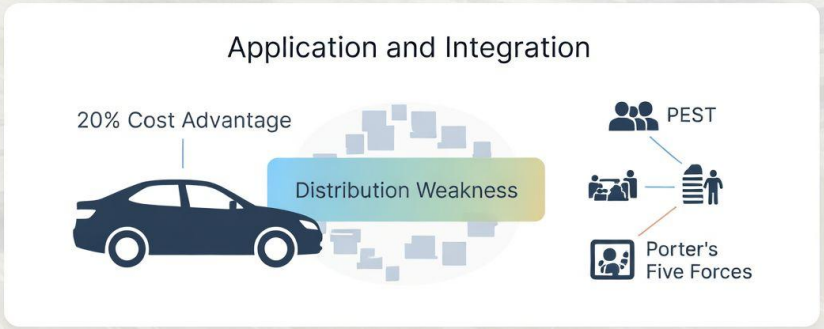
Powered by Poffices.AI

SWOT Analysis - Framework and Application









PEST Analysis: Political, Economic, Social, Technological Factors



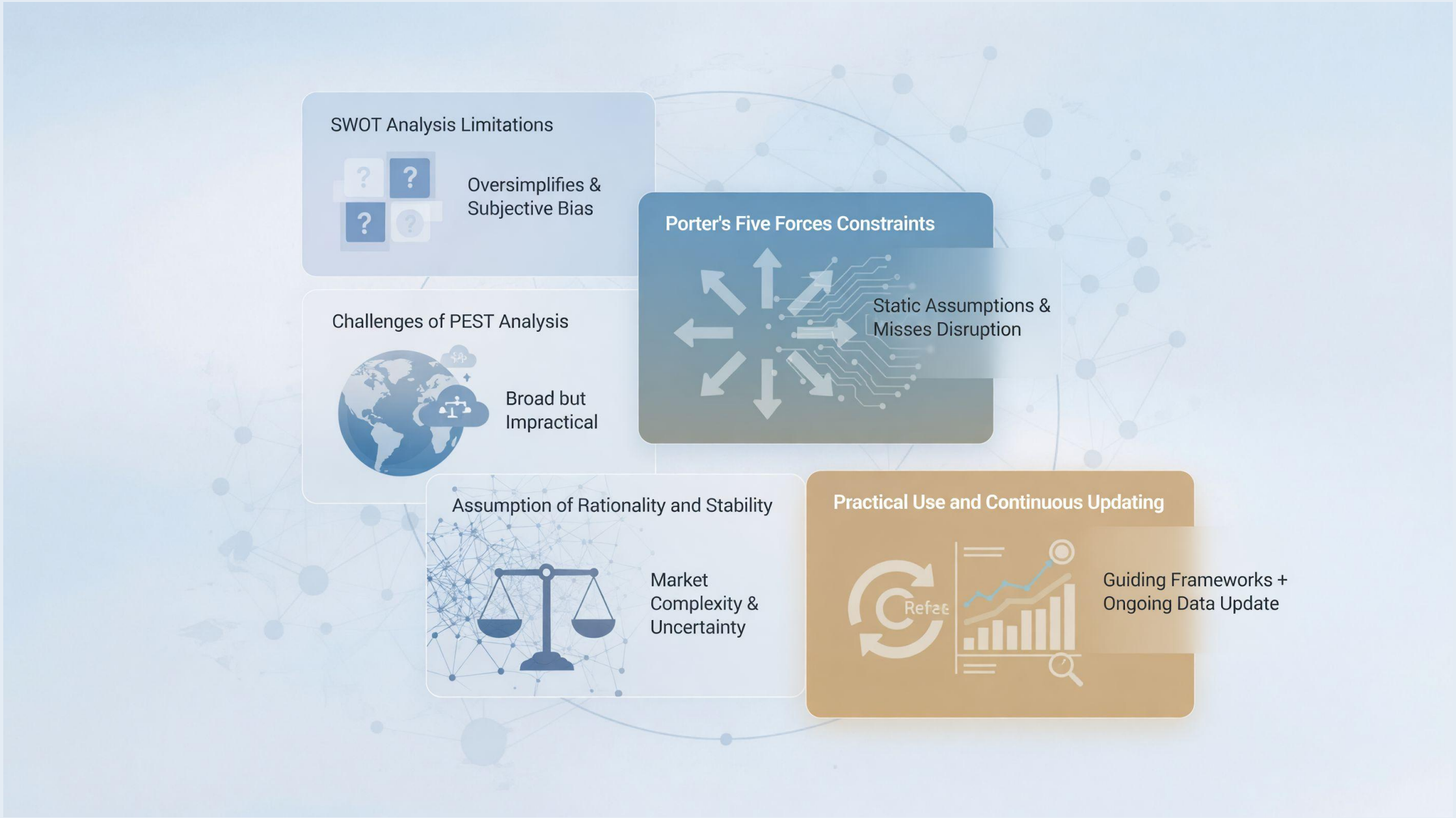
Porter’s Five Forces: Industry Competitive Forces



Market Scenarios Using Each Analysis Method

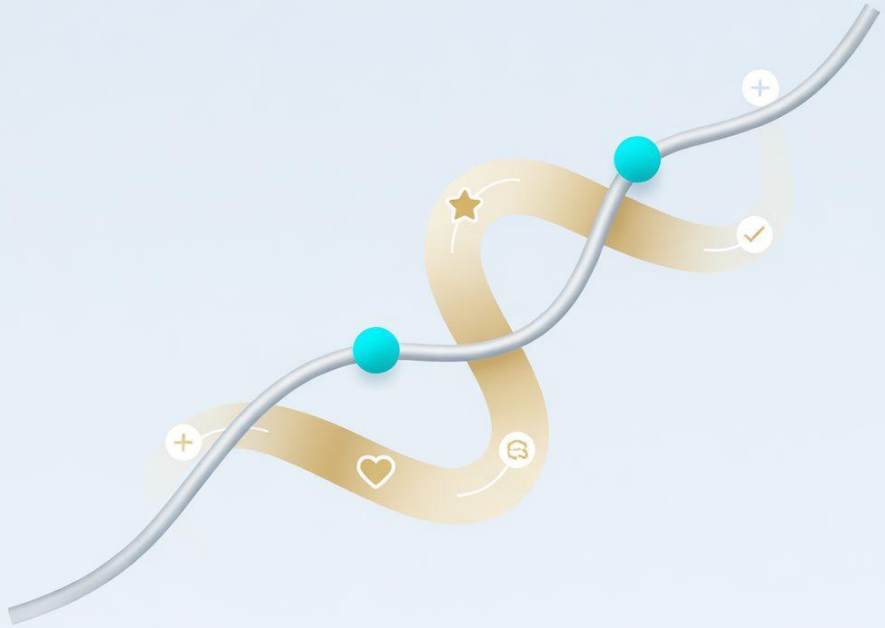


Limitations and Critical Evaluation of Market Analysis Methods



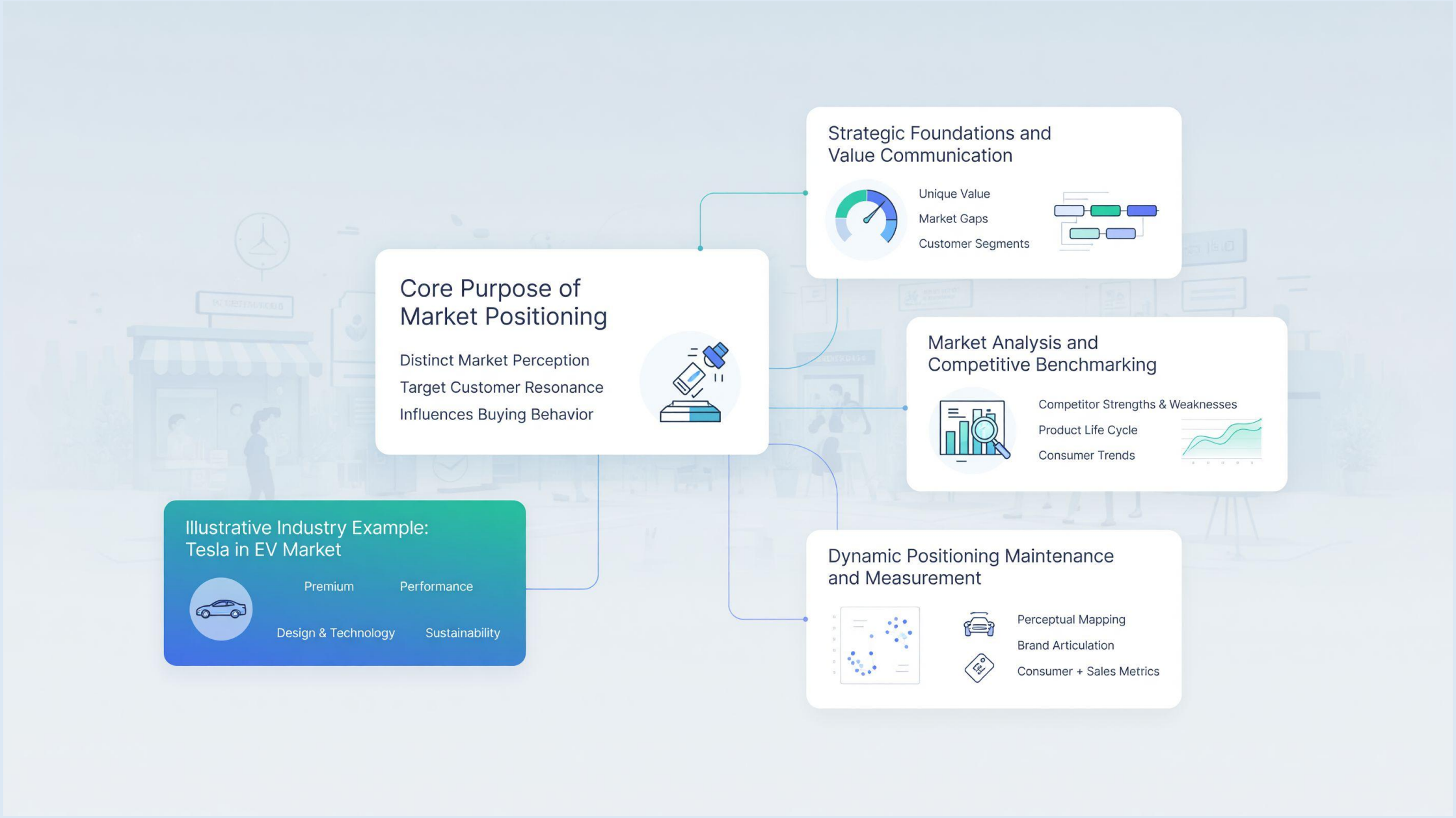
Chapter 5: Market Positioning of Products and Services (Kano Model and Customer Journey)

Ir Prof Alan Lam, JP



Powered by Poffices.AI

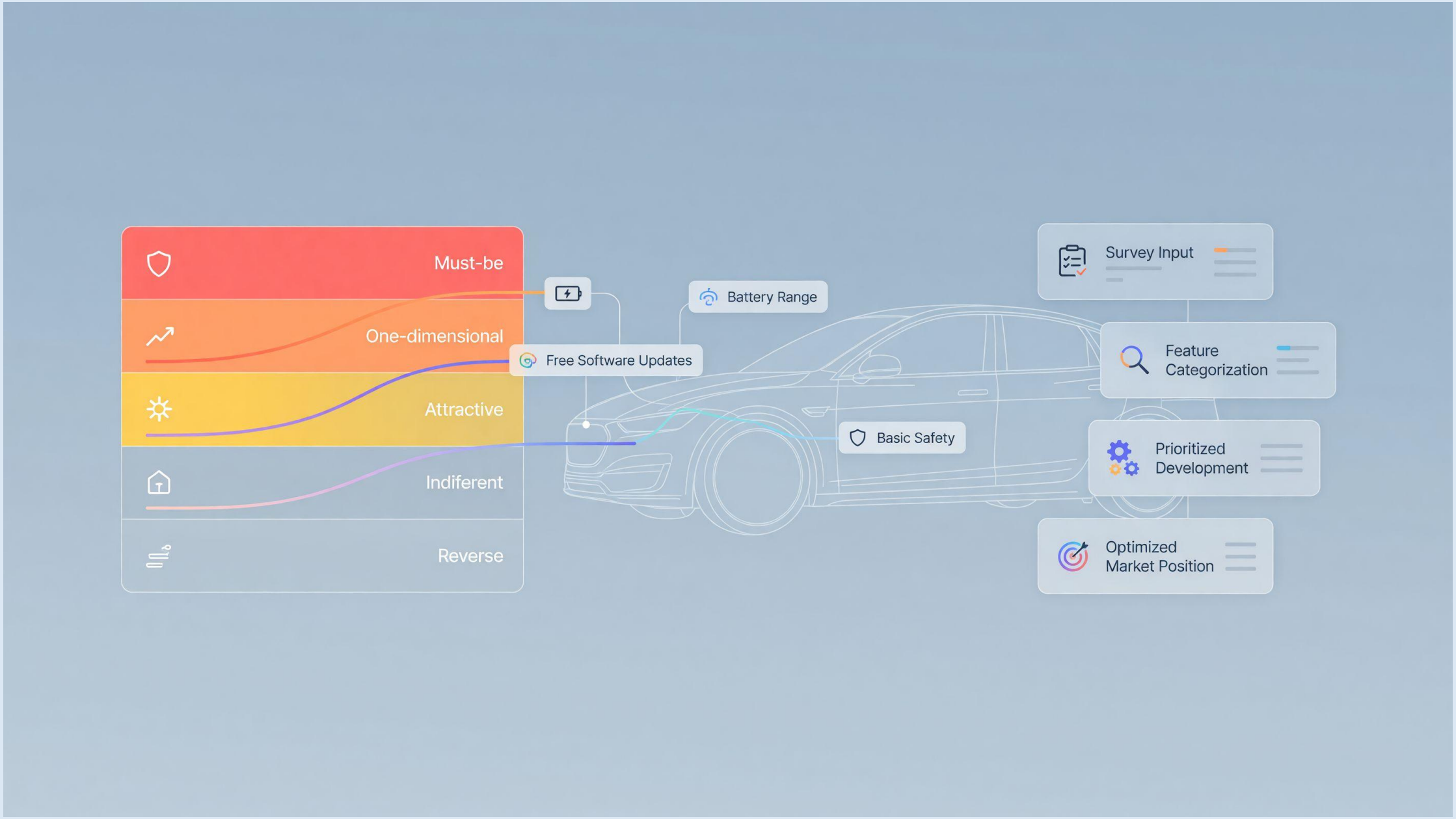
Fundamentals of Market Positioning



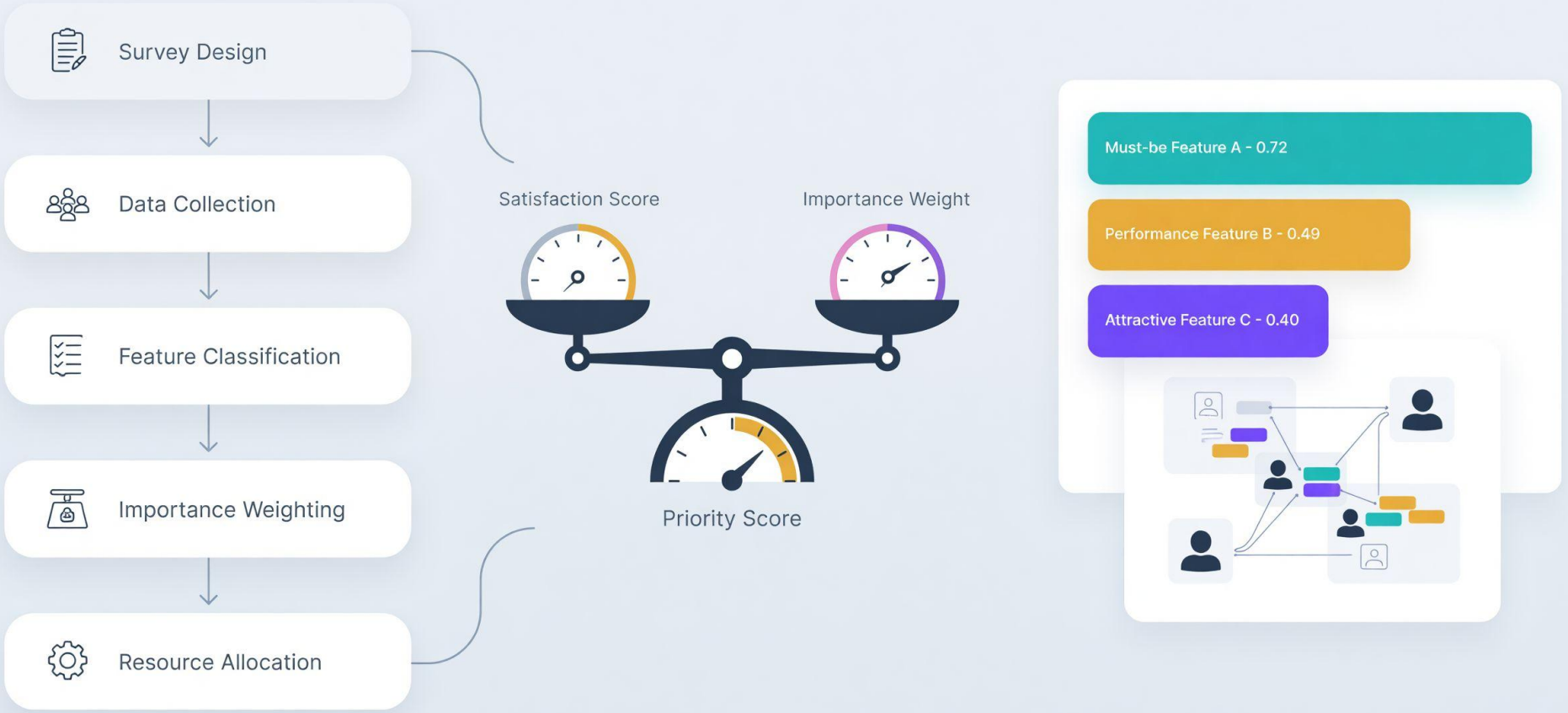
Kano Model: Categories and Customer Satisfaction



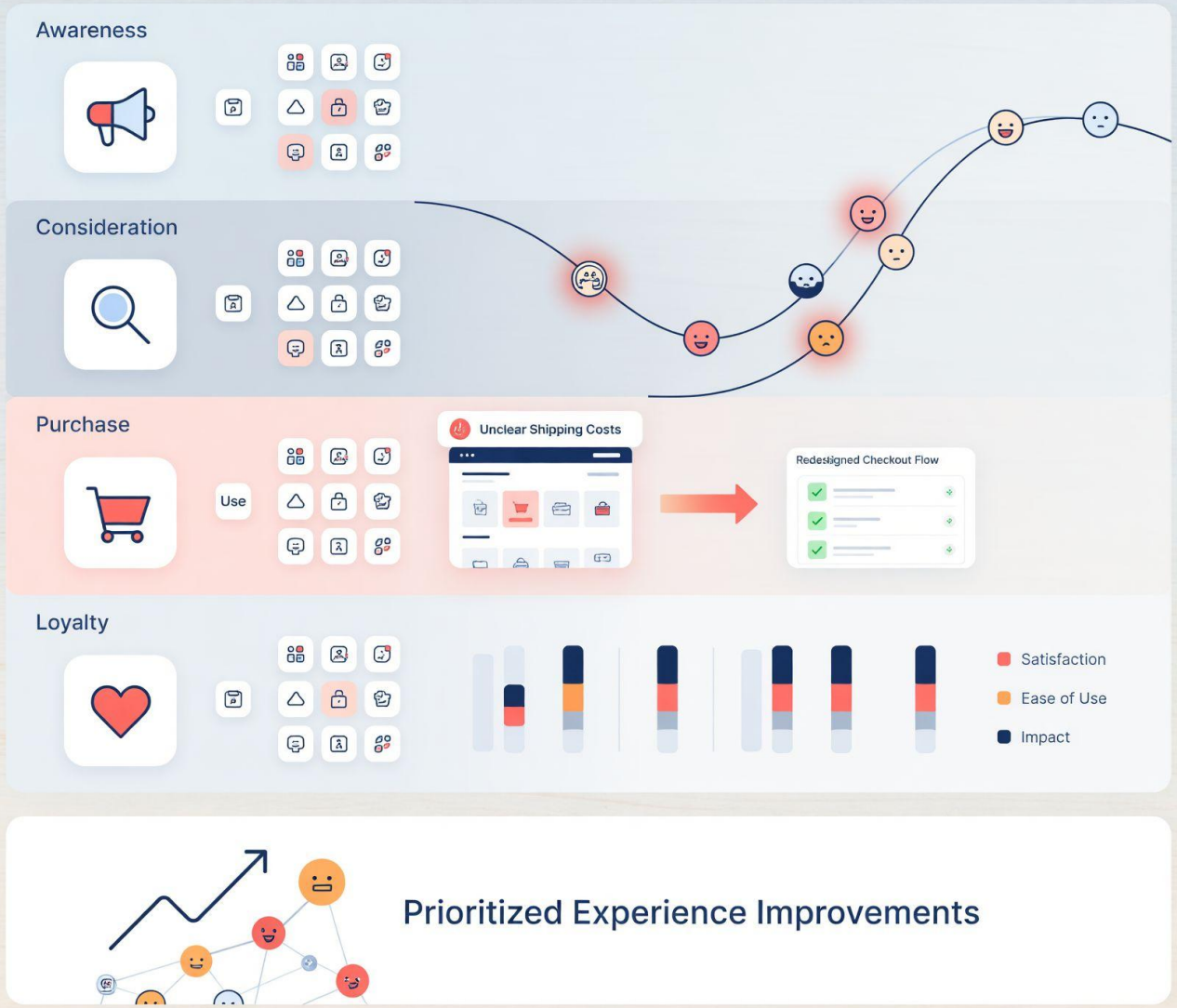
Kano Model Explanation and Application



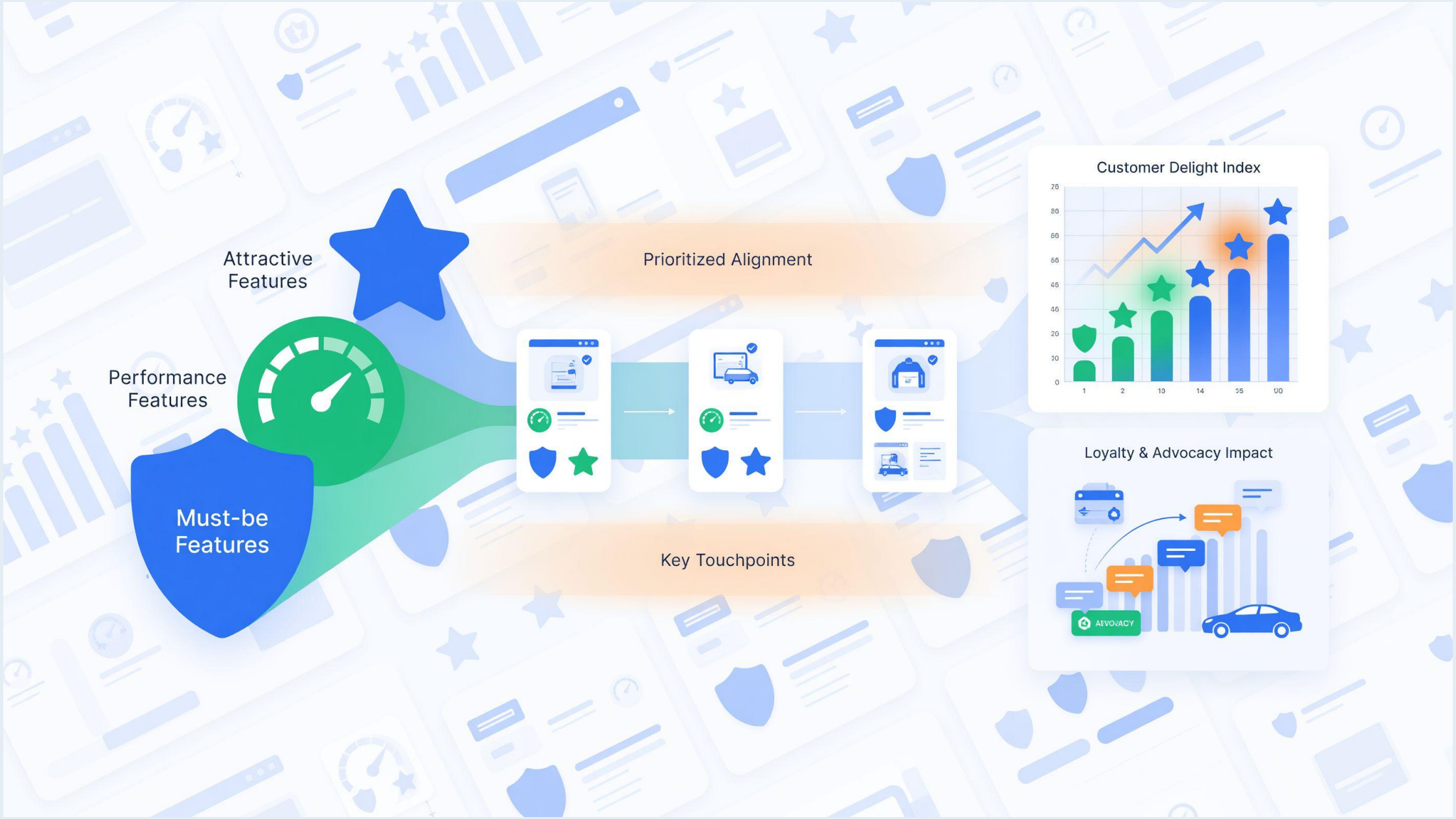
Using Kano Model for Product Feature Prioritization



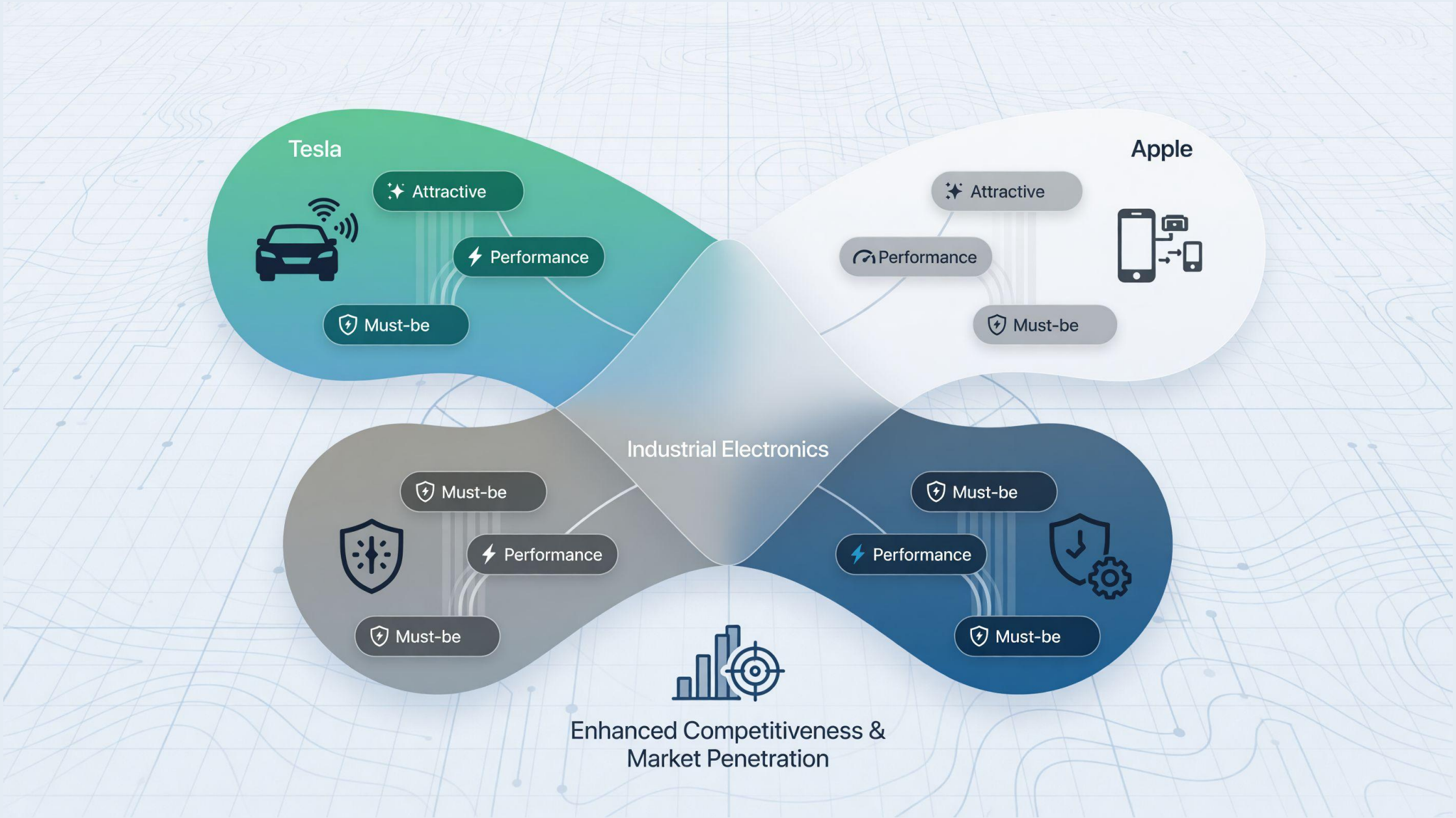
Customer Journey Mapping and Touchpoints



Linking Kano Model with Customer Experience



Examples of Market Positioning in Products



Chapter 6: Case Study of Tesla for Successful Product Identification

Ir Prof Alan Lam, JP



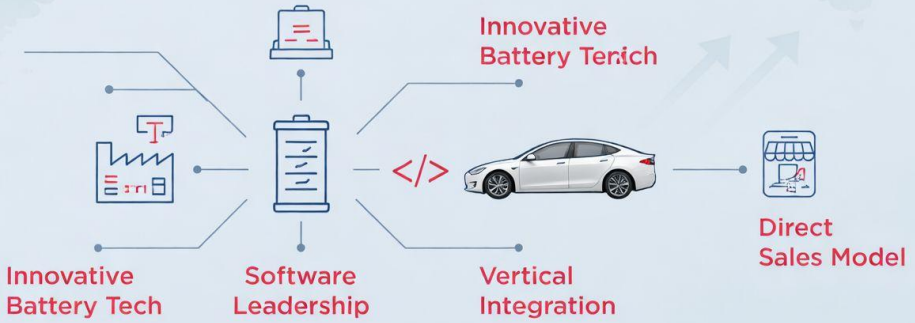
Powered by Poffices.AI

Tesla Company Background and Market Context





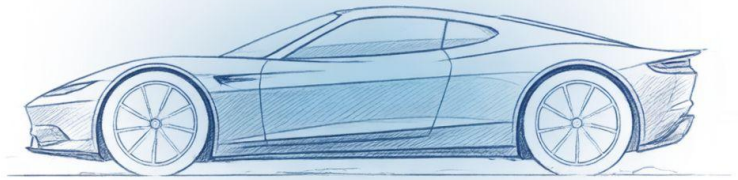
Tesla's Strategic Advantages and Market Opportunity



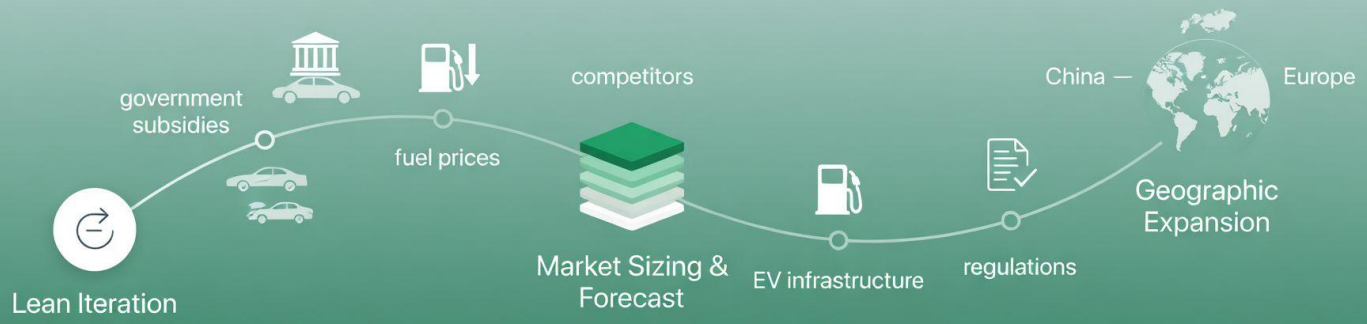
Tesla’s Market Opportunity Identification Process



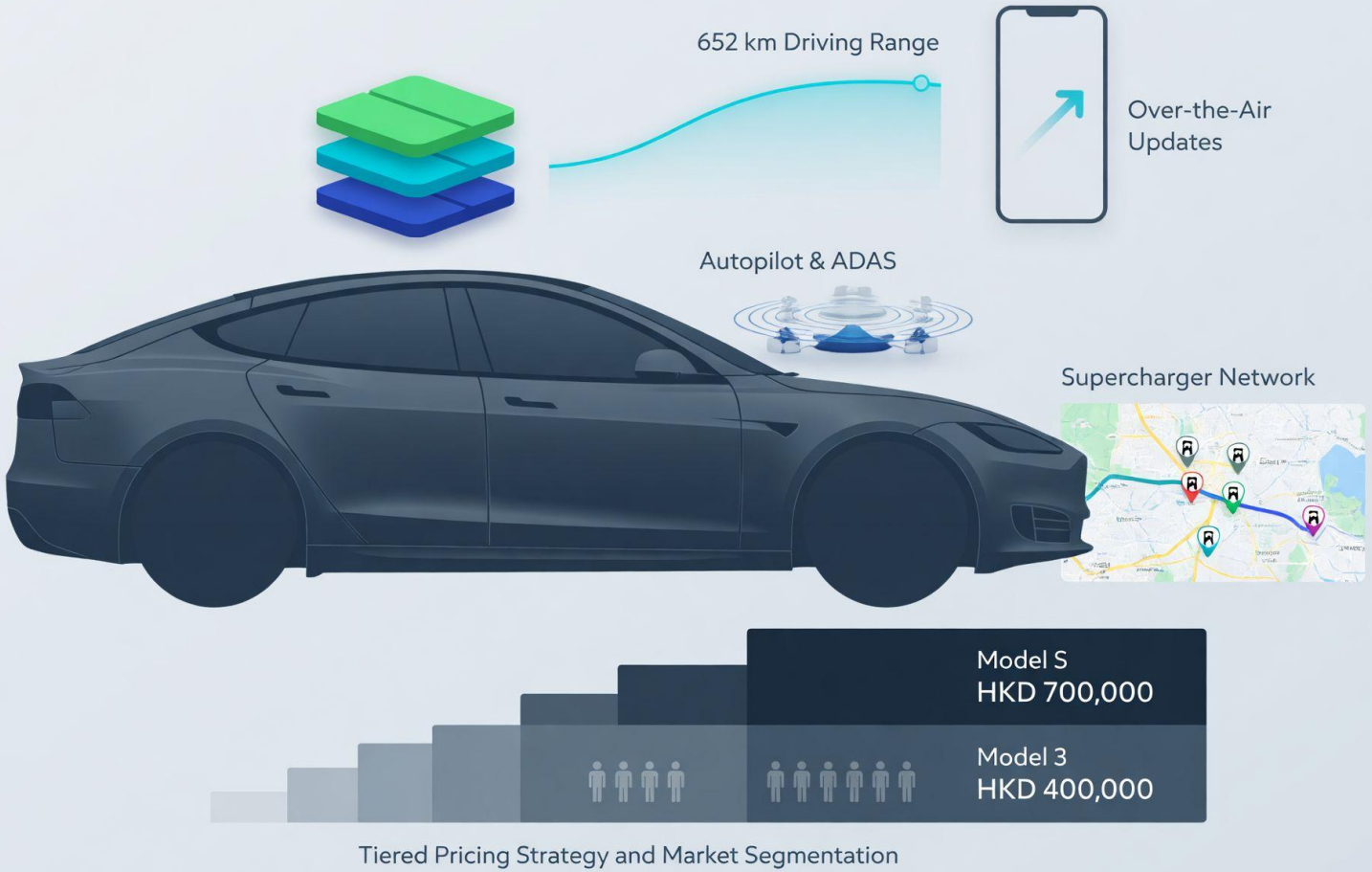
Premium EV Market Gap
Affluent Early Adopters



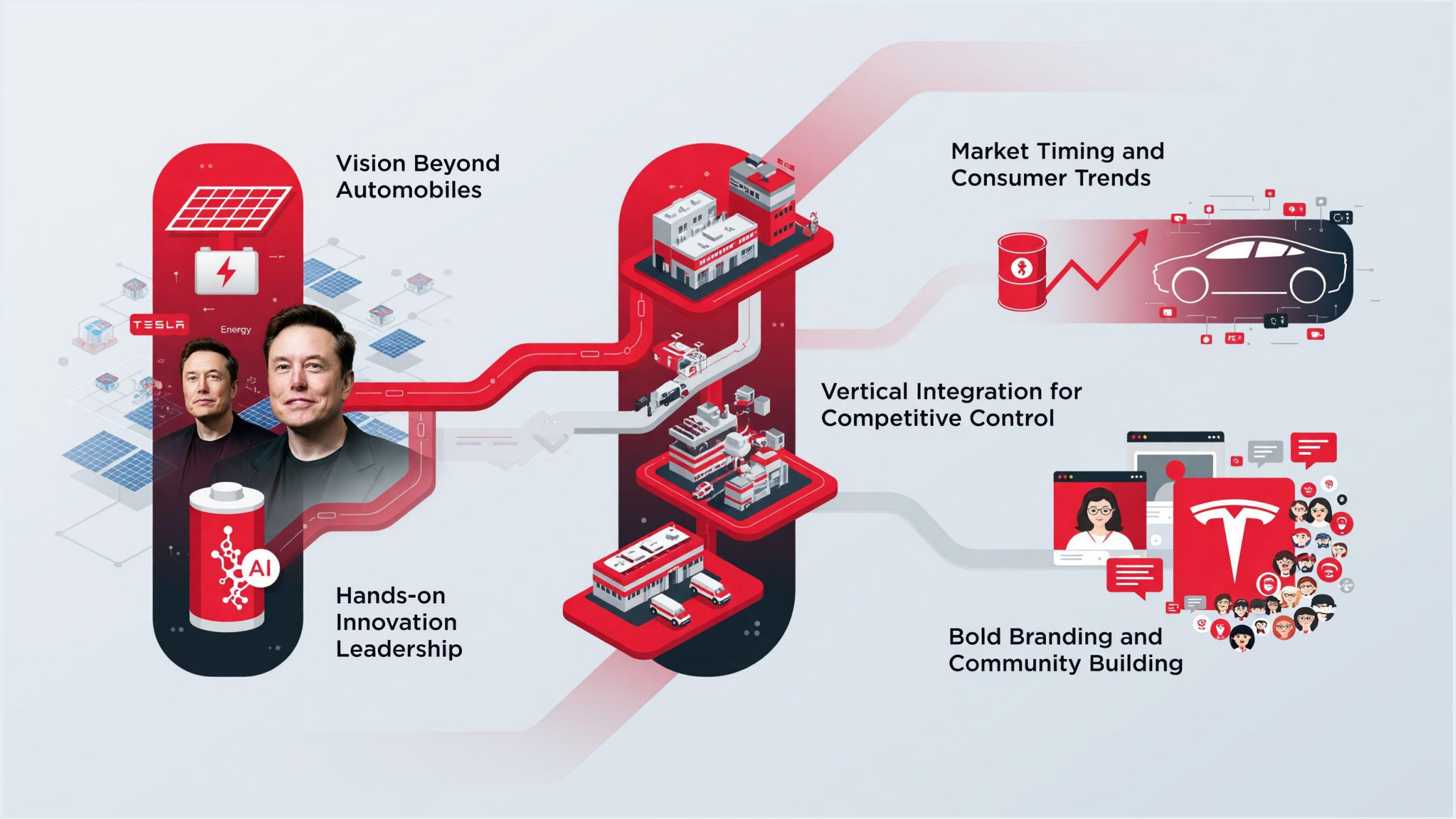
- Tech Innovation
- Market Validation
- Willingness to Pay



Innovative Product Features and Market Fit



Elon Musk’s Strategic Market Insights



Tesla’s Competitive Advantage Analysis



Challenges Faced and Solutions Implemented



Lessons from Tesla for Entrepreneurs



In-Class Discussion:

- Please based on the today's lecture to summarize what did you learn
- Based on the market identification methodologies to evaluate the market of your ideas about startup projects.
- Make one Slide about Problem Statement and Market Potential.



Market Potential Identification

Reference:

- 2025 Global Economy
- https://www.un.org/development/desa/dpad/publication/world-economic-situation-and-prospects-as-of-mid-2023/
- https://unstats.un.org/sdgs/report/2023/
- https://emeritus.org/blog/entrepreneurship-role-of-entrepreneurship-in-economic-development/
- https://www.agribusinessgroup.com/news/innovation-in-business-sustainability
- SMUHK_Leaflet_Survey_V4_rev 3 (startmeup.hk)
- Hong Kong Start-Up Survey: Ecosystem Overview | 香港貿易發展局經貿研究 (hktdc.com)



Ir Prof Alan Lam, JP
Adjunct Professor

End of Presentation
Thank You